AI Fraud Detection & Response Governance Framework

1. Executive Summary

This document presents an enterprise-grade governance and risk management framework for an AI-driven fraud detection and response system.

The framework extends beyond traditional fraud detection by incorporating a **risk-based response mechanism**, enabling controlled actions such as account freezing, provisional credit, and investigation workflows.

It integrates governance, risk, and operational processes to ensure:

- Effective fraud mitigation
- Customer protection
- Regulatory compliance
- Ethical AI usage

2. Business Context

Financial institutions face increasing fraud risks across digital channels such as:

- Online banking
- Mobile transactions
- Card payments

While fraud detection systems exist, **post-transaction response remains slow and inefficient**, leading to:

- Customer dissatisfaction
- Financial loss
- Reputational damage

This framework addresses that gap.

3. System Overview

AI Fraud Detection Engine

- Real-time transaction monitoring
- Behavioral analysis
- Anomaly detection

AI Fraud Response Mechanism

Trigger: Customer raises fraud complaint

System actions:

- AI re-evaluates transaction
- Assigns risk score
- Initiates appropriate response

4. Core Components

4.1 AI Fraud Detection Engine

- Detects suspicious transactions
- Uses machine learning models

4.2 Decision Engine

- Applies risk-based decisioning
- Determines action:
    - High Risk → Freeze + Provisional Credit
    - Medium Risk → Manual Review
    - Low Risk → Standard Process

4.3 Fraud Response Workflow

- Complaint intake
- AI re-evaluation
- Action execution
- Investigation
- Resolution

4.4 Governance Layer

- Board / Risk Committee
- AI Governance Council
- Risk & Compliance Teams
- Internal Audit

4.5 Monitoring & Feedback

- KPI tracking
- Model performance monitoring

- Continuous improvement

5. Risk Management Framework

Risk Scoring Model

- Likelihood (1–5)
- Impact (1–5)
- Risk Score = Likelihood × Impact

Key Risks

- Fraud not detected (false negatives)
- False positives impacting customers
- Abuse of complaint-based refund system
- AI bias and fairness issues
- Model drift
- Data privacy risks

6. Control Framework

Controls are implemented to mitigate identified risks:

- Threshold validation for decision engine
- Identity verification before action
- Approval limits for refunds
- Continuous monitoring and alerts
- Audit logging of AI decisions
- Periodic model review and retraining

7. Governance Model

The framework aligns with COBIT-based governance principles:

EDM (Evaluate, Direct, Monitor)

- Define AI risk appetite
- Provide oversight

APO (Align, Plan, Organize)

- Define policies and risk framework

BAI (Build, Acquire, Implement)

- Ensure model validation and testing

DSS (Deliver, Service, Support)

- Execute fraud detection and response

MEA (Monitor, Evaluate, Assess)

- Monitor performance and conduct audits

8. Fraud Response Workflow

1. Customer raises fraud complaint
2. AI system re-evaluates transaction
3. Decision engine assigns risk level
4. Action is executed:
    - Freeze account
    - Issue provisional credit
    - Initiate manual review
5. Fraud team investigates
6. Final resolution is completed
7. Feedback loop improves AI model

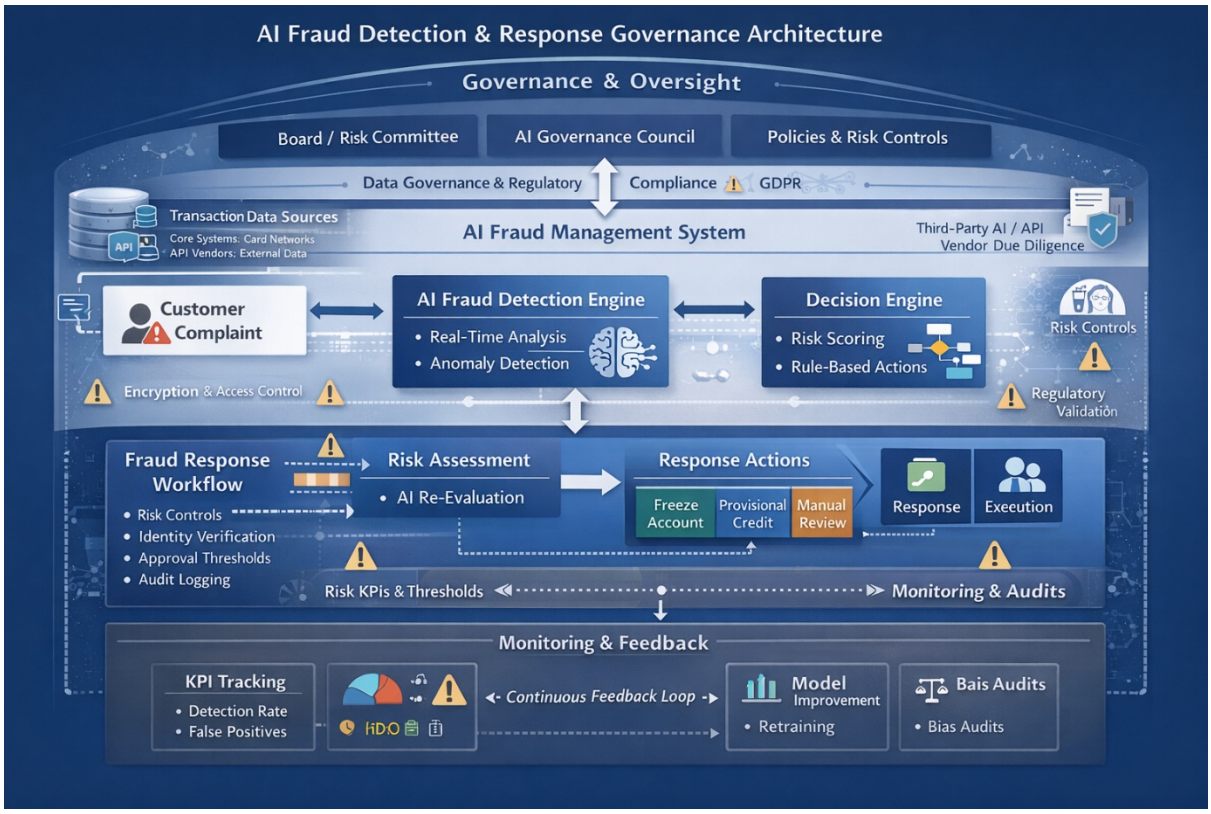
9. Key Metrics (KRIs)

- Fraud detection rate
- False positive rate
- Time to resolve fraud cases
- Number of provisional credits issued
- Recovery success rate

10. Conclusion

This framework establishes a **comprehensive, governance-driven AI fraud management system**, integrating detection, response, and risk control.

It enables:

- Faster and controlled fraud response
- Improved customer trust
- Strong governance and accountability
- Scalable AI adoption

Architecture Diagram